

ECHS FRAUD ANALYTICS

Geo-Spatial Fraud Clustering

MODULE 9 REPORT | FULL DATABASE COVERAGE | FY 2013 – FY 2025

PEAK REGION DEDUCTION

19.13% — Region 6

LARGEST REGION EXPOSURE

■840 Cr Deducted
Region 1

PARK HOSPITAL CHAIN

■369 Cr Deducted
5 Branches in Top 20

VIJAY HOSPITAL RATE

34.00% Deduction
Highest in Top 20

Region 6 — 19.13% Deduction

HIGHEST RISK

Park Hospital Chain — 5 Units

VERY HIGH RISK

Vijay / Madhuraj / Virat

HIGH RISK

IIT KANPUR — Data Analytics & Fraud Intelligence Division

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Executive Summary

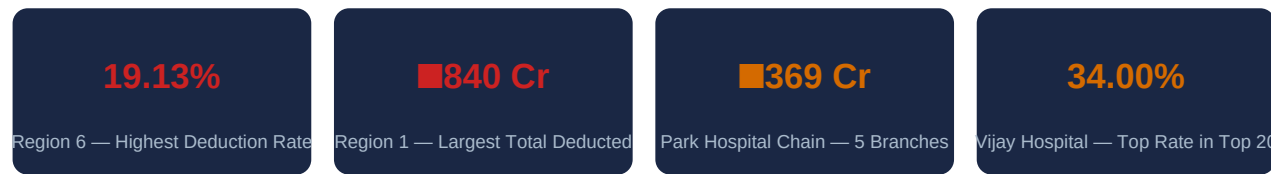
Module 9 analyses **geographic concentration of ECHS fraud** — identifying which regions have the highest deduction rates, which individual hospitals account for the largest absolute deduction amounts, and where multi-branch hospital chains are systematically overbilling across multiple locations.

Finding 1 — Region 6 is the highest-risk fraud geography at 19.13% deduction. Across 31 ECHS regions, Region 6 stands out with nearly one in five rupees claimed being deducted. Region 8 is the largest fraud exposure in absolute terms — **₹622 Cr** deducted from **₹4,127 Cr** claimed across 266 hospitals.

Finding 2 — The Park Hospital chain appears 5 times in the top 20 hospitals by deduction. Combined across its five empanelled units, Park Hospital has had **₹369.62 Cr deducted** — more than any other individual hospital or chain. Deduction rates across its units range from 13.44% to 28.55%, all significantly above the national average.

Finding 3 — Three hospitals have deduction rates above 29%. VIJAY HOSPITAL (34.00%), MADHURAJ HOSPITAL KANPUR (33.13%), and VIRAT HOSPITAL (29.99%) have deduction rates indicating that nearly one-third of everything they bill is being rejected by assessors — consistent with systematic overbilling rather than documentation errors.

Note on Q9b (Cross-State Claims): This query could not be executed. It requires joining **claim_intimation** to **office_master** via hospital ID, but the two tables use different identifier systems — **claim_intimation** uses text-based hospital codes while **office_master** uses numeric IDs. This analysis is reserved for local server access.



S1	Region 6 — 19.13% deduction, 119 hospitals, ₹1,129 Cr claimed	CRITICAL
S2	Park Hospital Gurgaon (367) — ₹172.54 Cr deducted, 22.95% rate	CRITICAL
S3	VIJAY HOSPITAL — 34.00% deduction rate on ₹440 Cr claimed	CRITICAL
S4	MADHURAJ HOSPITAL KANPUR — 33.13% rate on ₹124 Cr claimed	CRITICAL
S5	Region 8 — ₹622 Cr total deducted (largest absolute loss, 266 hospitals)	CRITICAL
S6	VIRAT HOSPITAL — 29.99% deduction rate on ₹155 Cr claimed	HIGH
S7	Park Hospital Kailash (1161) — 28.55% deduction, ₹48.65 Cr deducted	HIGH
S8	YATHARTH WELLNESS HOSPITAL — 24.26% rate, ₹46.43 Cr deducted	HIGH
S9	Region 13 — 15.78% deduction rate (second highest region)	HIGH
S10	Park Hospital chain total — ₹369.62 Cr deducted across 5 empanelled units	CRITICAL

Q9a

settlement_stat — all 31 ECHS regions

Regional Fraud Hotspot Analysis — Deduction Rates by Region

The query aggregates all claims in settlement_stat by ECHS region code (SS_REGION_ID), computing total claimed, approved, deducted, and deduction rate per region. Regions with high deduction rates represent geographic concentrations of overbilling — either due to a few dominant high-deduction hospitals or widespread billing manipulation across many facilities in that region.

Region	Hospitals	Claims	Claimed ■Cr	Approved ■Cr	Deducted ■Cr	Ded %
R-6	119	3.88L	■1,129	■913	■216	19.13%
R-13	79	4.57L	■647	■545	■102	15.78%
R-3	223	10.0L	■1,470	■1,243	■227	15.46%
R-25	83	4.86L	■569	■482	■87	15.31%
R-8	266	21.1L	■4,127	■3,504	■622	15.08%
R-20	72	1.06L	■172	■148	■24	14.05%
R-26	65	2.88L	■336	■290	■47	13.9%
R-11	61	2.76L	■267	■232	■34	12.92%
R-18	147	4.38L	■528	■460	■68	12.85%
R-5	140	8.59L	■1,255	■1,094	■161	12.83%
R-27	209	15.5L	■3,335	■2,914	■421	12.62%
R-16	73	1.42L	■205	■180	■25	12.12%
R-10	86	3.95L	■921	■812	■109	11.82%
R-17	141	8.78L	■1,221	■1,083	■138	11.3%
R-29	363	32.9L	■4,430	■3,933	■497	11.22%
R-1	429	42.8L	■7,905	■7,064	■841	10.63%
R-24	216	10.7L	■2,752	■2,465	■287	10.43%
R-32	30	2.91L	■375	■339	■36	9.65%
R-30	161	14.2L	■2,063	■1,870	■193	9.35%
R-14	57	2.40L	■274	■250	■25	8.98%
R-21	74	4.44L	■552	■502	■49	8.94%
R-19	124	5.63L	■688	■628	■61	8.82%
R-31	34	2.10L	■173	■158	■15	8.51%
R-28	84	4.67L	■764	■700	■64	8.37%
R-15	94	7.40L	■1,007	■923	■84	8.34%
R-2	116	30.8L	■1,911	■1,757	■154	8.06%
R-4	200	26.4L	■5,917	■5,490	■427	7.22%
R-33	51	3.43L	■306	■284	■22	7.07%
R-9	248	18.7L	■6,891	■6,407	■485	7.03%
R-7	133	24.4L	■2,106	■1,966	■141	6.67%
R-12	125	6.91L	■1,156	■1,081	■75	6.53%

Source: settlement_stat GROUP BY SS_REGION_ID. Deducted = Claimed – Approved. All 31 ECHS regions shown. Red = deduction rate ≥15% (critical). Orange = ≥12%. Region IDs are ECHS internal codes.

Key Findings — Regional Analysis

Region 6 has the highest deduction rate at 19.13% across 119 hospitals and █1,129 Cr in claims, with █216 Cr deducted. Regions 13 (15.78%), 3 (15.46%), 25 (15.31%), and 8 (15.08%) form a cluster of high-risk geographies all exceeding 15% deduction.

Region 1 and Region 8 represent the largest absolute fraud exposure. Region 1 has █840 Cr deducted from █7,905 Cr claimed (10.63% rate) — the largest total loss in the database by amount. Region 8 follows with █622 Cr deducted at a higher 15.08% rate. These two regions alone account for over █1,460 Cr in deductions.

Q9c

settlement_stat + office_master

Top 20 Hospitals by Total Deduction Amount — Priority Audit List

The query ranks all empanelled hospitals by total amount deducted across their entire history with ECHS. High total deductions indicate either very high claim volumes (which may be legitimate) or high deduction **rates** (which indicate overbilling). Both dimensions are shown. The Park Hospital chain appears **5 times** in the top 20 — the only chain with this level of concentration.

ID	Hospital Name	St.	Claims	Claimed ■Cr	Deducted ■Cr	Rate	Flag
367	PARK HOSPITAL — GURGAON (Park Medicentres)	8	1.23L	751.71	172.54	22.95%	PARK CHAIN
3149	VIJAY HOSPITAL	8	21.8K	440.20	149.66	34.00%	CRITICAL
1219	HEALING TOUCH SUPER SPECIALITY HOSPITAL	8	71.5K	548.75	85.87	15.65%	
60	PARK HOSPITAL — CHOWKHANDI	34	23.0K	290.58	66.80	22.99%	PARK CHAIN
183	MEDANTA THE MEDICITY — GURGAON	8	3.41L	564.34	49.12	8.70%	
1161	PARK HOSPITAL — KAILASH SUPERSPECIALITY	22	52.1K	170.41	48.65	28.55%	PARK CHAIN
109	GRECIAN SUPER-SPECIALITY HOSPITAL	30	50.2K	332.23	48.05	14.46%	
78	LIVASA HOSPITAL (IVY HEALTHCARE INFRA)	30	2.02L	579.54	46.96	8.10%	
2856	VIRAT HOSPITAL	8	22.2K	155.28	46.57	29.99%	CRITICAL
1454	YATHARTH WELLNESS HOSPITAL & TRAUMA CENTRE	26	1.36L	191.39	46.43	24.26%	
2127	PARK HOSPITAL — PARK MEDICITY	8	20.4K	201.07	43.29	21.53%	PARK CHAIN
1042	METRO HOSPITAL & HEART INSTITUTE — NOIDA	26	77.7K	268.39	43.25	16.11%	
399	MANIPAL HEALTH ENTERPRISES PVT LTD	22	3.18L	567.76	43.02	7.58%	
519	EMC SUPER SPECIALITY HOSPITAL	21	39.4K	405.04	42.28	10.44%	
34	FORTIS HOSPITAL — MOHALI	30	4.42L	809.00	41.47	5.13%	
292	MADHURAJ HOSPITAL — KANPUR	26	11.4K	124.12	41.12	33.13%	CRITICAL
514	PARK HOSPITAL — UMKAL HEALTHCARE	8	59.2K	285.23	38.34	13.44%	PARK CHAIN
17	ARTEMIS MEDICARE SERVICES — GURGAON	8	4.62L	628.60	36.99	5.88%	
1373	PARK HOSPITAL — FARIDABAD	8	19.5K	267.59	36.54	13.66%	PARK CHAIN
74	INDUS SUPER SPECIALITY HOSPITAL — MOHALI	30	99.7K	332.35	34.07	10.25%	

Source: settlement_stat LEFT JOIN office_master on SS_OFFICE_ID = OM_OFFICE_ID. Ranked by total SS_DED_AMT descending. St. = OM_OFFICE_STATE_ID. State 8 = Haryana, 30 = Punjab, 26 = Uttar Pradesh, 22 = Maharashtra.

Key Findings — Top Hospital Analysis

Park Hospital is the single most significant fraud concentration in the database. Five distinct legal entities operating under the Park Hospital brand appear in the top 20 — their combined deduction of **■369.62 Cr** dwarfs any other chain. All five have deduction rates above 13%, with the Kailash unit at 28.55%. The spread across multiple legal entities may be deliberate structuring to avoid triggering per-hospital audit thresholds.

Three hospitals have deduction rates above 29% — well above any defensible clinical justification. VIJAY HOSPITAL at 34.00% deduction on ■440 Cr, MADHURAJ HOSPITAL KANPUR at 33.13% on ■124 Cr, and VIRAT HOSPITAL at 29.99% on ■155 Cr. At these rates, assessors are consistently rejecting one-third of all claims — a pattern that indicates systematic overbilling, not isolated documentation issues.

State 8 (Haryana) dominates the top 20 by hospital count. Seven of the top 20 hospitals are in Haryana — primarily in the NCR/Gurgaon belt. This geographic concentration aligns with Region 6 being the highest-risk region.

Recommendations

1. EMERGENCY — Targeted audit of all 5 Park Hospital units simultaneously

The Park Hospital chain has ₹369.62 Cr in deductions across 5 legal entities. The distribution across multiple corporate vehicles (Park Medicentres, Park Medicity, Umkal Healthcare, Kailash Superspeciality) while maintaining a common brand suggests deliberate structuring. A coordinated audit across all 5 units should be conducted simultaneously to prevent evidence suppression. Office IDs for immediate action: 367, 60, 1161, 2127, 514.

2. IMMEDIATE — Recovery review for Vijay Hospital, Madhuraj, and Virat Hospital

VIJAY HOSPITAL (34.00%, ₹149.66 Cr deducted on ₹440 Cr claimed), MADHURAJ HOSPITAL KANPUR (33.13%, ₹41.12 Cr), and VIRAT HOSPITAL (29.99%, ₹46.57 Cr) have deduction rates that cannot be explained by documentation gaps alone. All amounts approved for these hospitals above a 15% deduction-rate threshold should be placed under recovery review.

3. HIGH — Field investigation in Region 6 (highest deduction rate, 19.13%)

Region 6 has the highest deduction rate across all 31 ECHS regions. A field audit team should be deployed to identify the specific hospitals within Region 6 driving this rate. The region contains 119 hospitals and ₹1,129 Cr in claims — the fraud is concentrated but systematic.

4. HIGH — Structural review of Regions 8 and 1 (largest absolute exposure)

Region 1 (₹840 Cr deducted) and Region 8 (₹622 Cr deducted) account for over ₹1,460 Cr of deductions combined. While their rates (10.63% and 15.08%) are lower than Region 6, their scale means even a 1% improvement in fraud detection accuracy represents over ₹100 Cr in savings annually.

5. PENDING — Q9b (Cross-State Claims) requires local server access

Query Q9b identifies beneficiaries from one state claiming at hospitals in a different state — a strong indicator of ghost claims or identity fraud. It requires joining claim_intimation and office_master, which uses incompatible identifier systems and cannot be executed over VPN. This analysis should be the first priority once database access is available on the local analysis server.